

EE 582 – Advanced Topics

(Energy Systems Modeling and Simulation)

Instructor:

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Class Webpage:

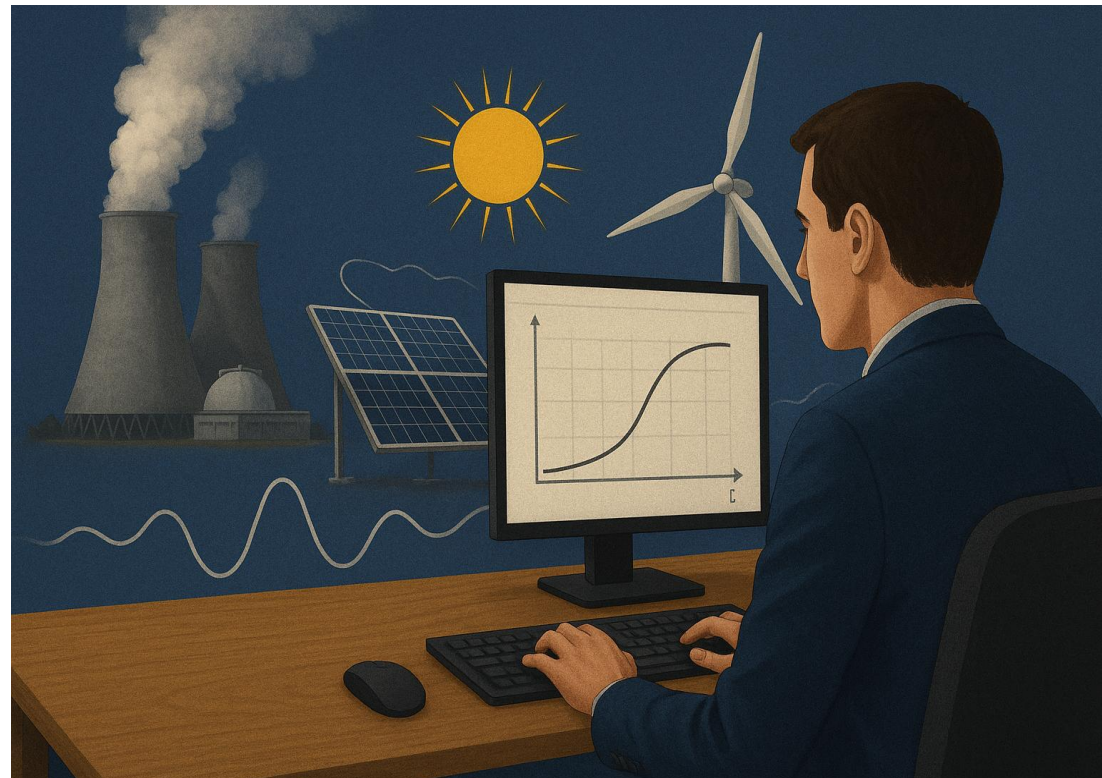
WSU Canvas

Lectures:

Tuesdays / Thursdays

4:20-5:35 pm

Fall 2025

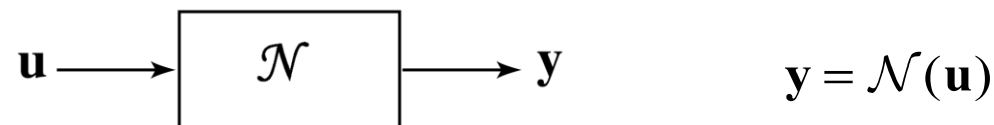


Purpose of Modeling

- ✓ **Scientists** study and analyze (usually) natural **systems**
- ✓ **Engineers** study and analyze (usually) man-made **systems**

□ What is a “System”?

A **system** is a collection of interacting elements, components, or subsystems that are deliberately **bounded** from their environment for the purpose of analysis, with well-defined **inputs**, **outputs**, **states**, and **dynamics** that describe its behavior over time



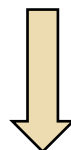
❖ Examples of Real Physical Systems:

Can range from very simple (e.g.,) to very complex systems (e.g., a generator, exciter, transmission line, converter, motor drive, load, etc.)

Purpose of Modeling

□ Why do we need *modeling*?

- It may be hard, impractical, or expensive to test the real physical system (and often many times!)



how to see what happens and then improve the design?!

- ✓ We often use the *model* of the system instead of the actual system itself, to analyze its behavior and conduct “**what if**” studies

□ What is a *model*?

- ✓ A mathematical (analytical, numerical, or combination of both) representation of the system, that is (typically) executed/simulated on a computer

□ What do we expect from a *model*?

- ✓ **The model is supposed to “mimic” the real system with the desired accuracy in some range of conditions**
- ✓ **What else? any restriction?!**

Approaches and Domains

➤ Models can be:

- Time invariant / Time-varying
- Lumped / Distributed parameter
- Linear / Nonlinear
- Static or steady-state
- Dynamic or transient
- In time, frequency, or phasor domain

Static Models: approximate input-output relationship **in steady-state**; typically they are described by algebraic equations

Dynamic Models: approximate input-output relationship **in steady-state and during transients**; **because they portray the system's dynamic behavior**, typically are described by differential and algebraic equations

Time domain: Describe how variables evolve with respect to time

Frequency domain: Represent signals or system behavior as a function of frequency rather than time

Phasor domain: Represent variables with sinusoidal steady-state approximation using complex numbers (phasors)

What is Time?



- Is it real and fundamental?
- Is it just an illusion emergent from quantum entanglement?
- Does it have only one direction/dimension?
- **Is it continuous or discrete in nature?**

Nature of Space/Time?

❖ Is Space/Time Continuous or Discrete?

➤ Classical View – Continuous

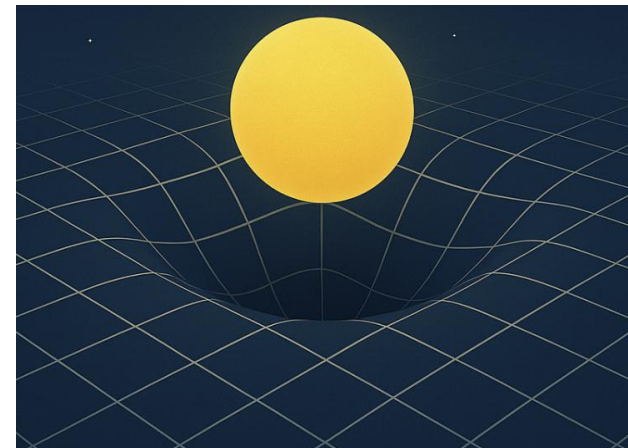
In **Einstein's relativity**, spacetime is modeled as a smooth, continuous four-dimensional fabric. So, you can always “zoom in” further into distances, trajectories, and also durations.

➤ Quantum View – Possibly Discrete

Quantum mechanics suggests that many physical quantities (like energy levels in atoms) are **quantized**, taking only discrete values. So, spacetime itself might also be quantized around the **Planck length/time**.

➤ Current Possible Measurements

Ultra-precise atomic clocks can detect time differences in the order of 10^{-19} seconds



- Planck length is approximately 1.6×10^{-35} meters, and Planck time is the time that light takes to travel a distance equal to the Planck length, i.e., approximately 5.4×10^{-44} seconds.

Nature of Space/Time?

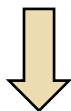
❖ Reality vs. Computer Simulations

➤ Reality (Physics):

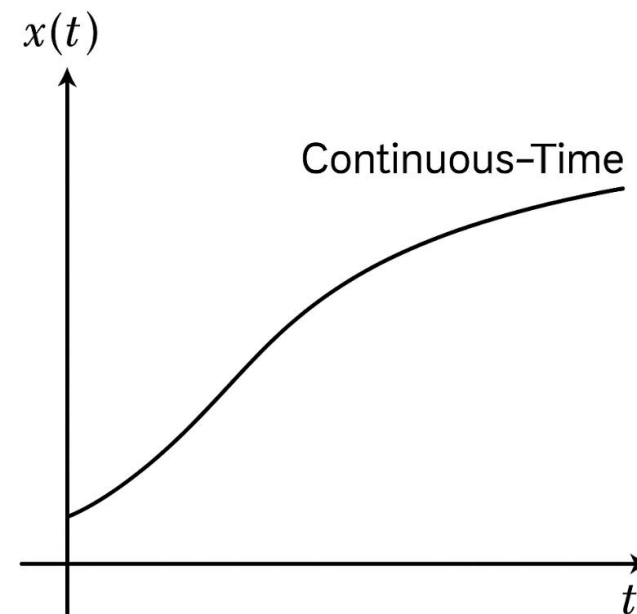
We do not really know whether time is fundamentally continuous or discrete. Let's also not worry about *time dilation*!

➤ Computer Simulations:

Regardless of the true nature of *time*, computers cannot handle *true continuity*. They must approximate by breaking time into finite steps, called **discretization**.

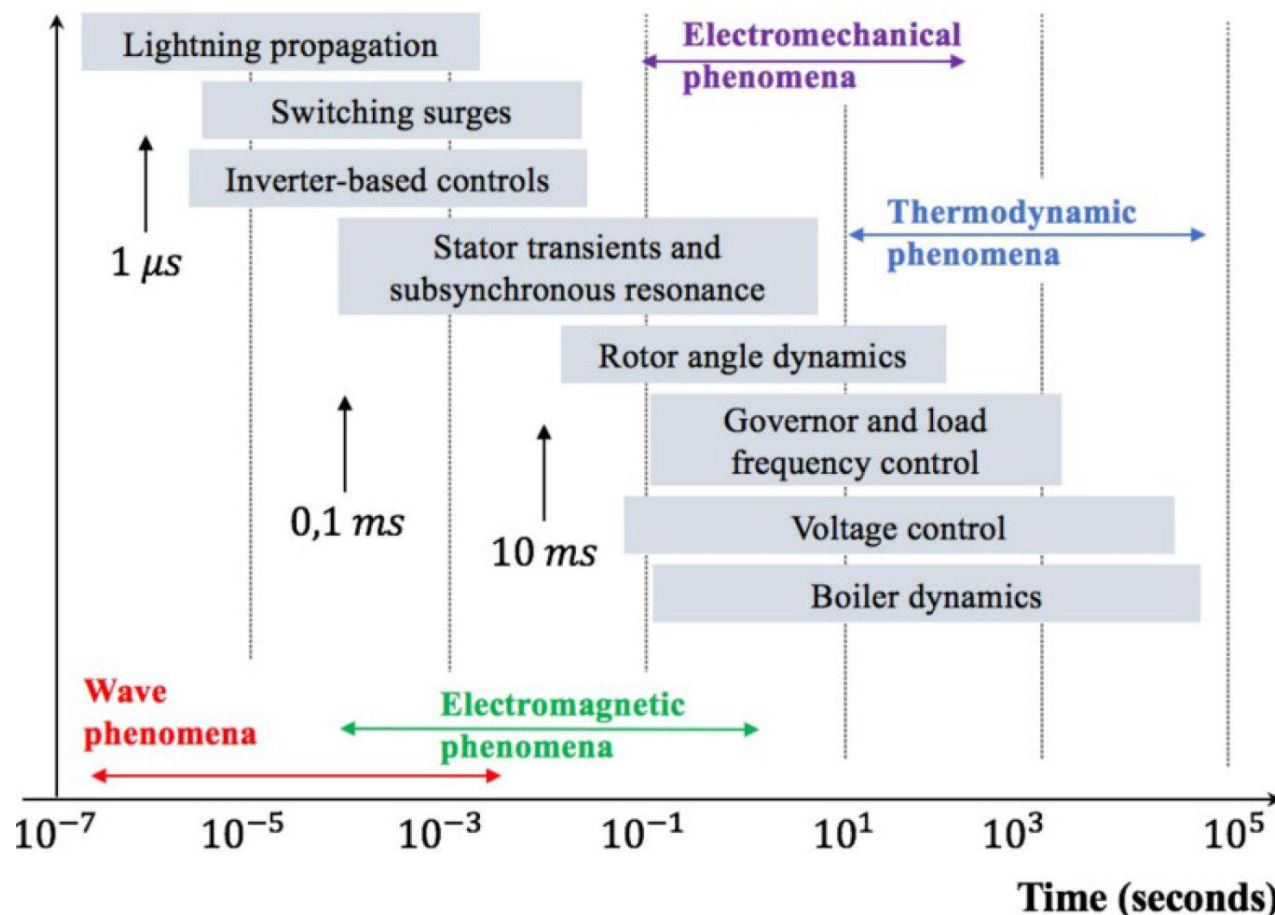


➤ How to choose time-step size?



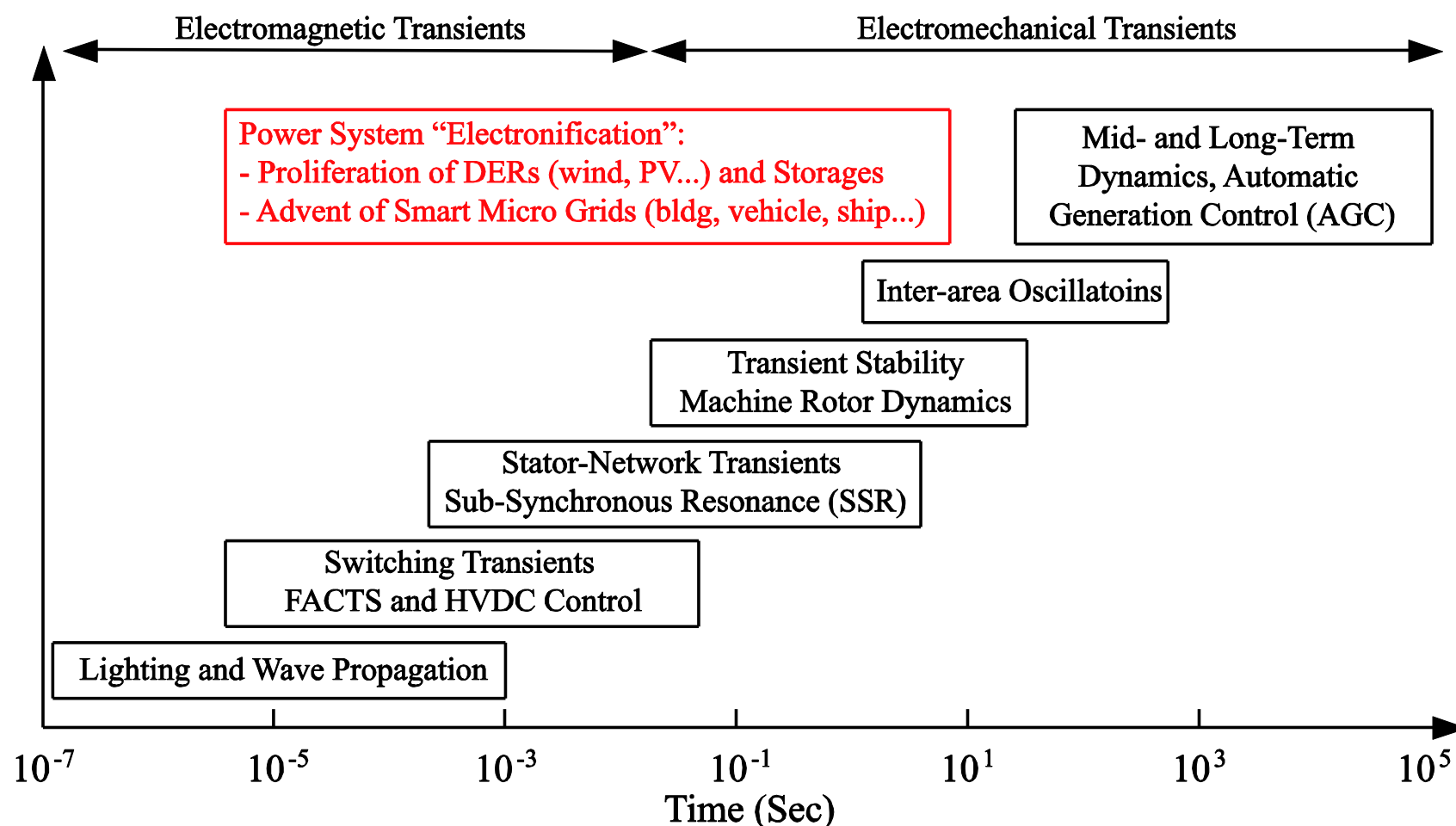
Time Scales of Phenomena in Power Systems

- ✓ In power systems, phenomena are classified by the time scale of their effect on physical processes, control actions, and system conditions

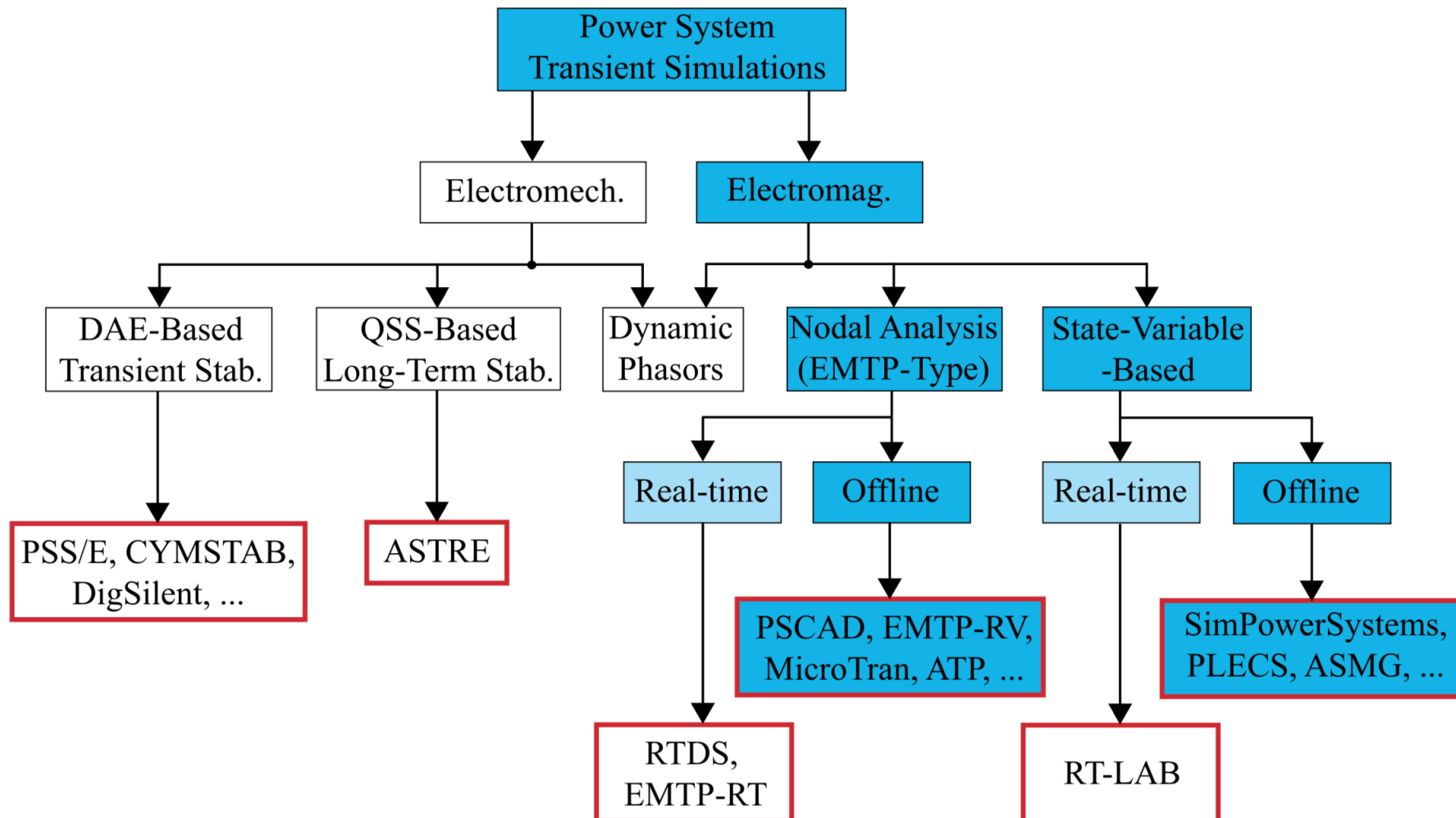


Time Scales of Phenomena in Power Systems

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Simulation Tools for Power Systems



- There can be other simulation programs which may not have been listed here.

Types of Tools and Simulation Methods for Power System Studies

❑ RMS-Type (Positive-Sequence):

- **Method:** Represents voltages and currents as RMS phasors at the fundamental frequency
- **Simplifications:** Assumes balanced systems (positive-sequence only)
- **Time Step Size:** Large (milliseconds)
- **Applications:** Transient stability, long-term stability, planning
- **Examples:** PSS®E, GE PSLF, PowerFactory (RMS mode)

❑ Phasor-Type (Per-Phase):

- **Method:** Per-phase phasors, not just positive-sequence
- **Simplification:** Can handle unbalance/asymmetry, but still not instantaneous waveforms
- **Time Step Size:** Milliseconds to sub-milliseconds
- **Applications:** Fault analysis, DG studies, unbalanced conditions
- **Examples:** PowerFactory (three phase mode), ETAP, OpenDSS

❑ Dynamic Phasor (DP):

- **Method:** Represents signals as slowly varying Fourier coefficients (generalized phasors)
- **Strength:** Captures harmonics and dynamic frequency content without full EMT
- **Time Step Size:** Larger than EMT but smaller than RMS
- **Applications:** Harmonic interactions, converter dynamics, multi-frequency systems
- **Examples:** Mostly implemented in MATLAB/Simulink, Opal-RT's ePHASORSim

Types of Tools and Simulation Methods for Power System Studies

❑ EMT (Electromagnetic Transients, Instantaneous Time-Domain):

- **Method:** Solves differential equations for instantaneous voltages and currents (waveforms)
- **Detail:** Captures switching, harmonics, fast control loops, resonance
- **Time Step Size:** Microseconds
- **Applications:** Power electronics, converter-grid interaction, protection studies
- **Examples:** PSCAD/EMTDC, EMTP, RTDS, Simulink Simscape Electrical, Hypersim

❑ Hybrid RMS–EMT (Co-Simulation):

- **Method:** Combines RMS for the bulk system with EMT for detailed subsystems (e.g., an inverter)
- **Motivation:** Efficiency + accuracy (zoom-in on critical subsystems)
- **Time Step Size:** Mixed (microseconds in EMT zone, milliseconds in RMS)
- **Applications:** IBR integration, grid-forming converters, HVDC interaction studies
- **Examples:** PSCAD–PSS®E co-sim, PSCAD-TSAT

Simulation Tools for Power Systems

➤ Differential Algebraic

- ✓ PSS/E (Siemens/PTI), PSLF (GE), DSATools (PowerTech Labs), DIgSILENT PowerFactory (DIgSILENT GmbH), CYME (Eaton), etc.

➤ Dynamic Phasors

- ✓ ePHASORsim

➤ Nodal-analysis-based (EMTP-type)

- ✓ MicroTran (Univ. of British Columbia)
- ✓ EMTP (BPA: Bonneville Power Administration)
- ✓ EMTP-RV, Hypersim (Hydro Quebec)
- ✓ PSCAD/EMTDC (Manitoba Hydro Int., Univ. of Manitoba)
- ✓ RSCAD (RTDS Technologies Inc.)
- ✓ SPICE/PSPICE (MicroSim Corporation; U of California, Berkeley; Cadence Design Sys.)
- ✓ ATP-EMTP (free academic variant of EMTP, EEUG Association)
- ✓ XTAP (CRIEPI: Central Research Institute of Electric Power Industry, Japan)

➤ State-variable-based

- ✓ Simscape Electrical toolbox in Simulink (Hydro Quebec)
- ✓ PLECS (ETH, Swiss Federal Institute of Technology, Plexim GmbH)
- ✓ ASMG (PCKA/Purdue Univ.)
- ✓ RT-Lab (Opal-RT Technologies Inc.)
- ✓ Typhoon-HIL (real-time simulator)

Simulation Tools for Power Systems

❖ Real-World Events where Lack of EMT Modeling Led to Issues:

- Several well-documented grid events illustrate the consequences of relying only on phasor-domain stability studies:

➤ South Australia Blackout (2016):

A major wind-penetrated system collapsed when multiple wind farms tripped due to control/protection interactions with voltage dips. Traditional RMS studies had suggested adequate ride-through capability, but EMT studies later showed the detailed converter control instabilities that were missed.

➤ California (CAISO) PV Plant Disturbances (2017):

Multiple large solar farms tripped simultaneously during a transmission fault. RMS-based models failed to reproduce the plant-level controller behavior. EMT simulations, incorporating manufacturer-provided controls, successfully replicated the event.

➤ Texas Grid Disturbances (2021, 2022):

Several ERCOT reports highlight unexpected IBR tripping during grid disturbances. EMT studies revealed fast frequency and voltage control interactions that RMS models had not captured.

Simulation Tools for Power Systems

❖ Real-World Events where Lack of EMT Modeling Led to Issues:

- **Odessa Event, Texas (2022):** Many solar plants tripped offline across a very large geographic region due to a fault near Odessa
- ✓ Report issued by North American Electric Reliability Corporation (NERC) concluded that many of the causes could not be sufficiently studied using traditional positive-sequence simulation tools; and recommended an increase in the use of EMT-level models and tools, to capture the dynamics presented by the IBRs more accurately.

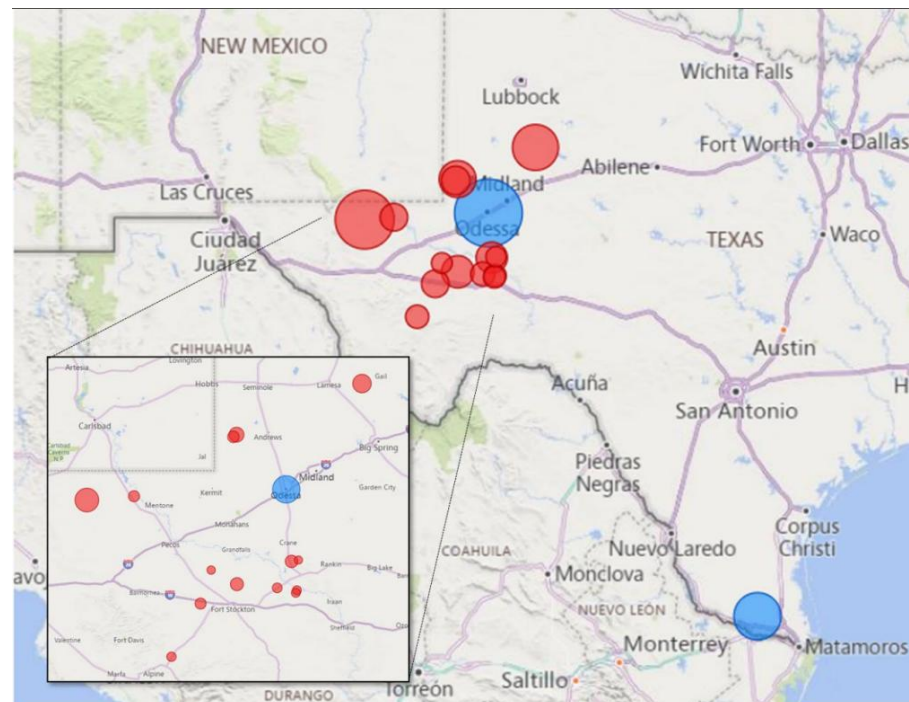
NERC
NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

2022 Odessa Disturbance

Texas Event: June 4, 2022

Joint NERC and Texas RE Staff Report

December 2022



Map of solar plants (red) and other generation (blue) that went offline or stopped producing due to the disturbance reported in the NERC 2022 Odessa Disturbance

- Not doing EMT simulation can lead to major prediction failures, with significant economic/reliability impacts. 15

Simulation Tools for Power Systems

❖ Importance of EMT Simulations:

Phasor-domain or positive-sequence transient stability (TS) simulation tools are numerically efficient, but they increasingly fall short in today's grid due to:

➤ High Penetration of Inverter-Based Resources (IBRs):

PV, wind, and battery systems are controlled by power electronics, which interact at sub-cycle dynamics. EMT models capture switching, phase imbalances, and control interactions that phasor models may miss

➤ Fast Dynamics and Control Interactions:

Converter control loops operate in the (sub) millisecond range where RMS models (with ~10–20 ms resolution) cannot predict the interactions

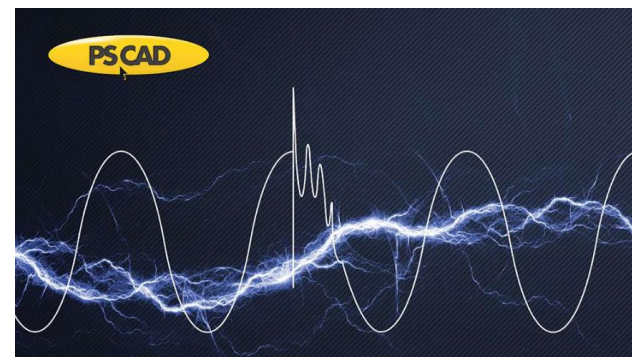
➤ Protection and Fault Ride-Through:

EMT is needed to model DC offsets, harmonic distortion, zero-sequence currents, and detailed breaker/protection responses

✓ EMT simulation is indispensable for **stability, reliability, and protection studies in converter-dominated grids**

Simulation Tools for Power Systems

- The primary tools currently used by the industry to study these phenomena are the commercial EMT software packages tailored for power systems transients (e.g., PSCAD)



❑ Developing Efficient EMT Simulation Tools is a Hot Research Area

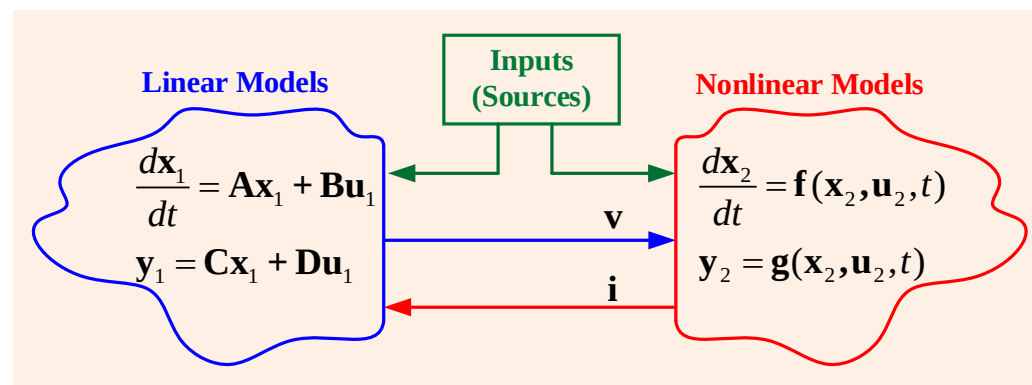
- ✓ The Electric Power Research Institute (EPRI) maintains an open-source power system solver (OpenDSS) that has recently been expanded to include dynamic simulation
- ✓ The Hierarchical Engine for Large-scale Infrastructure Co-Simulation (HELICS) open-source platform was recently developed under DOE's Grid Modernization Laboratory Consortium funding
- ✓ Pacific Northwest National Laboratory (PNNL) has also developed a platform to enable the use of parallel high-performance computing platforms for power system simulations, called GridPACK™

Simulation Tools for Power Systems

❖ State-Variable-Based Programs:

- ✓ Based on state-space model of the system

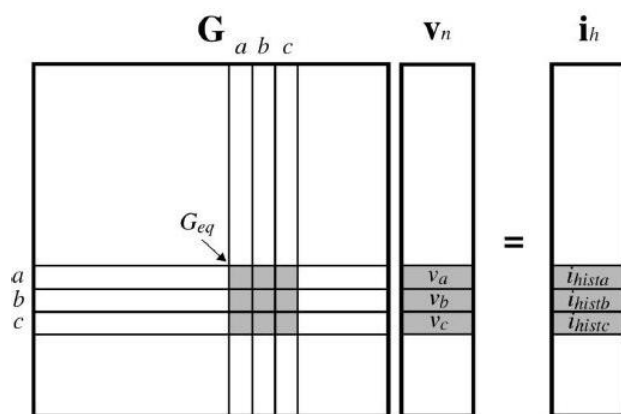
$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \\ \mathbf{y} = \mathbf{g}(\mathbf{x}, \mathbf{u}, t) \end{cases}$$
- ✓ The program keeps track of the **state variables**, not directly the node voltages



❖ Nodal-Analysis-Based or EMTP-type Programs:

- ✓ Solve directly for the node voltages based on KCL equations for each node

$$\mathbf{G}\mathbf{v}_n = \mathbf{i}_h$$



(a) Continuous Time	$i_{mk}(t) = \frac{1}{R} v_{mk}(t)$	$i_{mk}(t) = \frac{1}{L} \int v_{mk}(t) dt$	$i_{mk}(t) = C \frac{dv_{mk}(t)}{dt}$
(b) Discrete Time	$i_{mk}(t) = g_R v_{mk}(t)$ $g_R = \frac{1}{R}$	$i_{mk}(t) + i_{hmk} = g_L v_{mk}(t)$ $\begin{cases} g_L = \frac{\Delta t}{2L} \\ i_{hmk} = -[g_L v_{mk}(t - \Delta t) + i_{mk}(t - \Delta t)] \end{cases}$	$i_{mk}(t) + i_{hmk} = g_c v_{mk}(t)$ $\begin{cases} g_c = \frac{2C}{\Delta t} \\ i_{hmk} = [g_c v_{mk}(t - \Delta t) + i_{mk}(t - \Delta t)] \end{cases}$
(c) Discrete Time Equivalent Circuit			

Simulation Tools for Power Systems

- ✓ The foundation of all EMTP-type programs was established by Prof. Dommel mostly while at Bonneville Power Administration (BPA) in 1960s.



**Known as the
father of EMTP!**

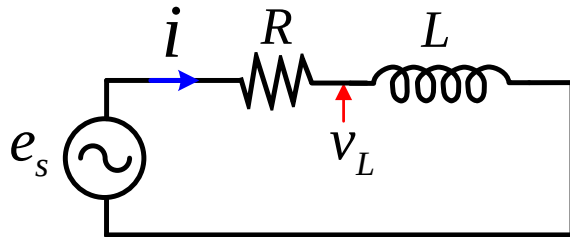


Prof. Hermann W. Dommel
(December 1933 – July 2025)

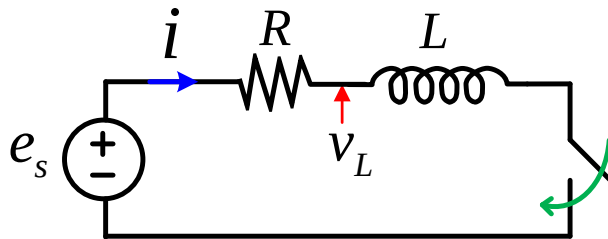
- He was with The University of British Columbia (UBC) in Vancouver, BC, Canada, since 1973.

Electric Circuits Solution

❖ Phasor Analysis:



❖ Transient Analysis:



❖ Analytical Solution:

❖ Numerical Step-by-Step Solution:
